

Live Lecture – Week 2

- Overall context (where we are)
- Revising last week
- Context for this week
- Preview for next week
- Q&A

Documents

Queries

1. Preprocessing

2. Index Construction

Index

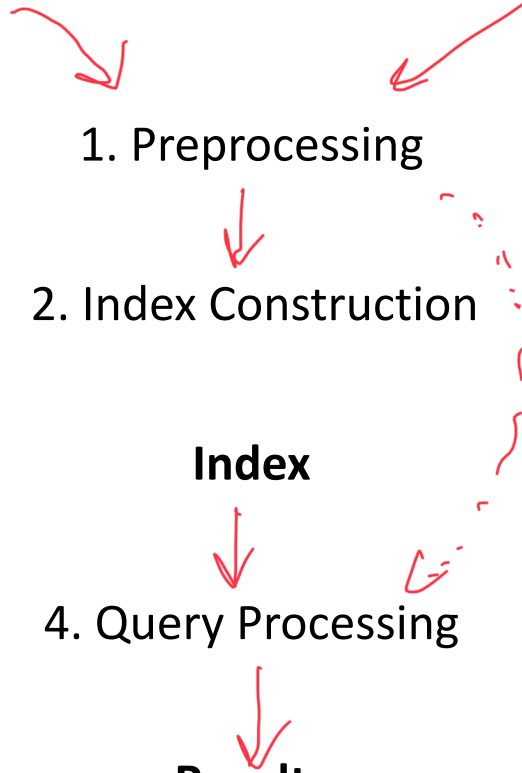
4. Query Processing

Results

5. Evaluation

3. Compression

0. Models



Web Search

- New Challenges
- How to obtain data?
- Additional features for Web data

6. Web Crawling

7. Link Analysis

	Relevant	Nonrelevant
Retrieved	tp	fp
Not Retrieved	fn	tn

- Precision $P = tp / (tp + fp)$
- Recall $R = tp / (tp + fn)$

TP: relevant results correctly retrieved

FP: non-relevant results incorrectly retrieved

TN: non-relevant results correctly not retrieved

FN: relevant results incorrectly not retrieved

	Relevant	Nonrelevant
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- Precision $P = tp / (tp + fp)$
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There are 80 documents containing a keyword *Apple*.
A search on *Apple* returns 60 documents.
Of the 60 documents retrieved, 45 contains *Apple*.

	Relevant	Nonrelevant
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There are 80 documents containing a keyword *Apple*.

A search on *Apple* returns 60 documents.

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Precision =

Recall =

	Relevant	Nonrelevant
Retrieved	tp	fp
Not Retrieved	fn	tn

- Precision $P = tp / (tp + fp)$
- Recall $R = tp / (tp + fn)$

There are 80 documents containing a keyword *Apple*.

A search on *Apple* returns 60 documents.

Of the 60 documents retrieved, 45 contains *Apple*.

Precision = $45 / 60 = 0.75$ or 75%

Recall = $45 / 80 = 0.56$ or 56%

	Relevant	Nonrelevant
Retrieved	tp	fp
Not Retrieved	fn	tn

- Precision $P = tp / (tp + fp)$
- Recall $R = tp / (tp + fn)$

There are 80 documents containing a keyword *Apple*.

A search on *Apple* returns 60 documents.

Of the 60 documents retrieved, 45 contains *Apple*.

Precision = $45 / (45+15) = 0.75$ or 75%

Recall = $45 / (45+35) = 0.56$ or 56%

TP = 45

FP = $60 - 45 = 15$

FN = $80 - 45 = 35$

TN?

Query processing: AND

- Consider processing the query:

Brutus AND Caesar

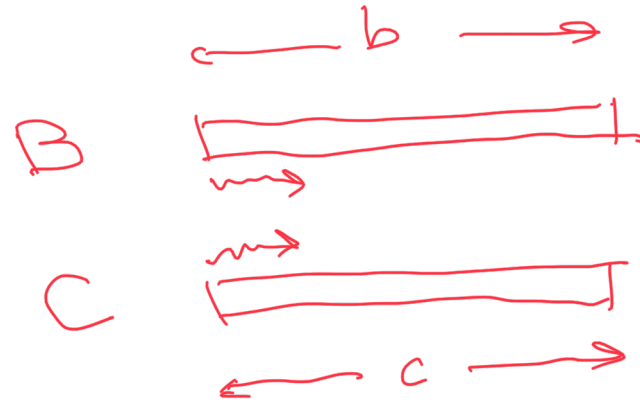
- Locate ***Brutus*** in the Dictionary;
 - Retrieve its postings.
- Locate ***Caesar*** in the Dictionary;
 - Retrieve its postings.
- “Merge” the two postings:

Query processing: AND

- Consider processing the query:

Brutus AND Caesar

- Locate ***Brutus*** in the Dictionary;
 - Retrieve its postings.
- Locate ***Caesar*** in the Dictionary;
 - Retrieve its postings.
- “Merge” the two postings:



$$O(b+c)$$

Query optimization example

- Process in order of increasing freq:
 - *start with the smallest set, then keep cutting further.*

This is why we kept document freq. in dictionary

Brutus	b	→	2	4	8	16	32	64	128	
Caesar	c	→	1	2	3	5	8	16	21	34
Calpurnia	p	→	13	16						

Execute the query as (**Calpurnia AND Brutus**) AND **Caesar**.

If we compute $c \& b$ first, then

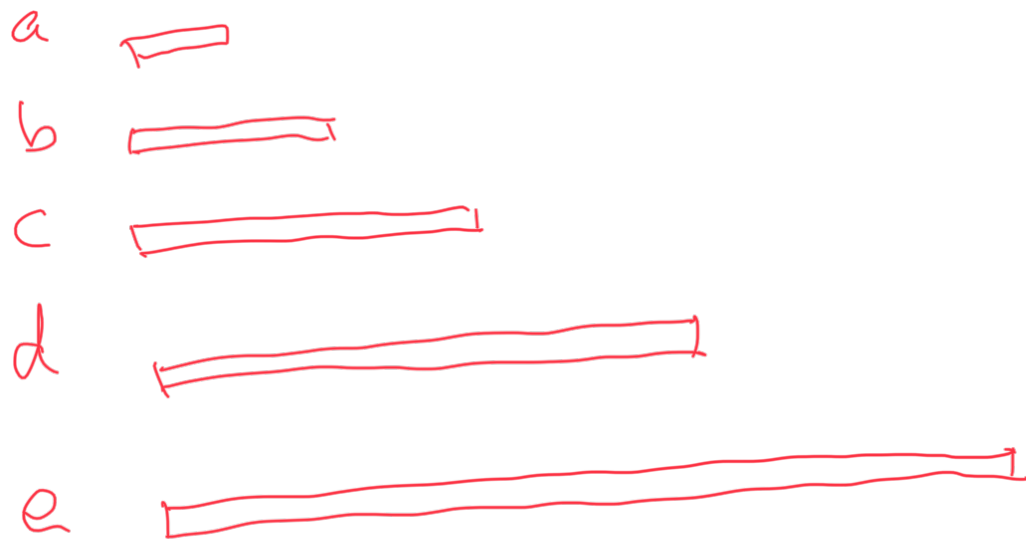
$$O(c + b + p + \min(b, c)) \quad ?$$

What about $p \& c$ first ?

$$1. O(p + b)$$

$$2. O(\min(p, b) + c)$$

$$O(c + b + p + \min(p, b))$$



$\min(a, b)$

a:

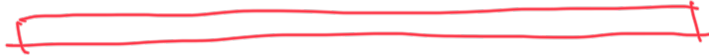
3	17
---	----

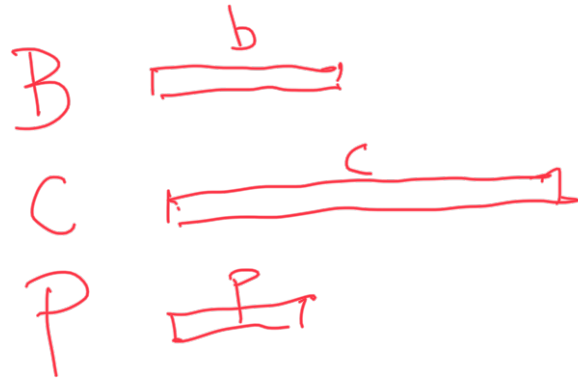
b: 

$\min(a, e)$

a:

3	17
---	----

e: 



B OR P OR C

1. $O(b + p)$
2. $O((b + p) + c)$

$$O(2(b + p) + c)$$

$$O(b + p + c)$$

Preview for this week:
Tokenization & normalization –
language dependent

Dictionary entries

<i>ensemble.french</i>
<i>時間.japanese</i>
<i>MIT.english</i>
<i>mit.german</i>
<i>guaranteed.english</i>
<i>entries.english</i>
<i>sometimes.english</i>
<i>tokenization.english</i>



Mit Apple



Language dependent

All

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About 519,000,000 results (0.46 seconds)

<https://ist.mit.edu> › hardware › apple

Recommended Computers: Apple - MIT.IST

MIT fully supports the use of **Apple** technology and is here to assist students, parents, faculty, and staff purchase technology that will help you succeed ...

<http://kb.mit.edu> › confluence › display › istcontrib › S...

Student Computer Purchase Discount Links - IS&T Contributions

21 June 2021 — **Apple**, Dell and Lenovo repair information is found at <https://ist.mit.edu/mail-in-hardware-repair> . IS&T Contributions. Documentation and ...

<https://apps.apple.com> › app › mit-app-inventor

MIT App Inventor on the App Store

MIT App Inventor is an educational tool to learn computational thinking and computational action principles through building mobile apps.

★ ★ ★ ★ ★ Rating: 2 · 350 reviews · Free · iOS · Educational

<https://apps.apple.com> › app › mit-atlas

MIT Atlas on the App Store

Read reviews, compare customer ratings, see screenshots, and learn more about MIT Atlas.

Download MIT Atlas and enjoy it on your iPhone, iPad, and iPod touch.

★ ★ ★ ★ ★ Rating: 3.2 · 180 reviews · Free · iOS · Educational

<https://www.technologyreview.com> › silicon-valley › a...

Apple | MIT Technology Review

The iPhones, Macs, and iPads made by **Apple**, the company created by Steve Jobs and now led by Tim Cook, have had more influence on our technology habits than ...

About 49,100,000 results (0.49 seconds)

Recipes



Küchlein mit Apfel

LIDL Kochen

4.2 ★★★★★ (31)

St, el



Apfel-Zimt-Küchlein

Chefkoch

4.7 ★★★★★ (158)

45 mins

Butter



Sauerkraut mit Apfel

GuteKueche.at

4.6 ★★★★★ (232)

50 mins

Sauerkraut, apfel

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<https://www.gutekueche.at/apfel-re...> · [Translate this page](#)

Die besten Apfel Rezepte | GuteKueche.at

Ein Apple Crumble ist ähnlich wie ein Streuselkuchen. Das Rezept wird mit **Äpfel** und einer Zucker-Zimt-Mischung zubereitet. oesterreichischer-apfelstrudel.



<https://www.chefkoch.de/apfel/Re...> · [Translate this page](#)

Apfel Rezepte - Chefkoch

Apfel - Wir haben 34.851 leckere **Apfel** Rezepte für dich gefunden! Finde was du suchst - erstklassig & einfach. Jetzt ausprobieren mit ♥ Chefkoch.de ♥.





cat



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About 6,050,000,000 results (0.72 seconds)

<https://en.wikipedia.org/wiki/Cat>

Cat - Wikipedia

The **cat** (*Felis catus*) is a domestic species of small carnivorous mammal. ... It is the only domesticated species in the family Felidae and is often referred to as ...

Kingdom: [Animalia](#)

Order: [Carnivora](#)

Genus: [Felis](#)

Phylum: [Chordata](#)

[Cat communication](#) · [Feral cat](#) · [Farm cat](#) · [Leopard cat](#)



<https://www.cat.com>

Cat.com | Cat | Caterpillar

Welcome! Select a region and language to continue. Select a Region, Africa, Middle-East, Asia, Australia, New Zealand, Eurasia, Europe, Latin ...

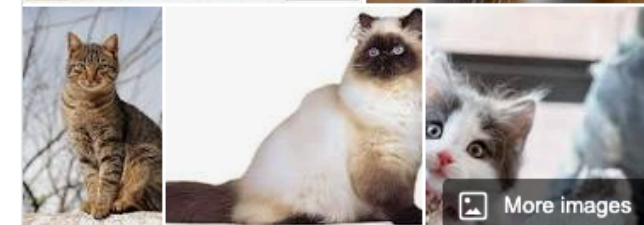


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[Nyan Cat \[original\]](#)

YouTube · Nyan Cat
6 Apr 2011



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Cat

Animal



The cat is a domestic species of small carnivorous mammal. It is the only domesticated species in the family Felidae and is often referred to as the domestic cat to distinguish it from the wild members of the family. [Wikipedia](#)



C.A.T.



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About 6,170,000,000 results (0.71 seconds)

<https://en.wikipedia.org/wiki/Cat>

Cat - Wikipedia

The **cat** (*Felis catus*) is a domestic species of small carnivorous mammal. ... It is the only domesticated species in the family Felidae and is often referred to as ...

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Genus: [Felis](#)

Phylum: [Chordata](#)

[Cat communication](#) · [Feral cat](#) · [Farm cat](#) · [Leopard cat](#)



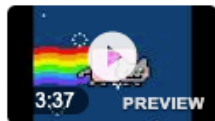
<https://www.cat.com>

Cat.com | Cat | Caterpillar

Welcome! Select a region and language to continue. Select a Region, Africa, Middle-East, Asia, Australia, New Zealand, Eurasia, Europe, Latin ...



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6 Apr 2011



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Cat

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C.A.T. profit

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Caterpillar revenue for the twelve months ending June 30, 2022 was **\$54.031B**, a 17.74% increase year-over-year. Caterpillar annual revenue for 2021 was \$50.971B, a 22.09% increase from 2020. Caterpillar annual revenue for 2020 was \$41.748B, a 22.4% decline from 2019.

<https://www.macrotrends.net> › stocks › charts › revenue

Caterpillar Revenue 2010-2022 | CAT - Macrotrends

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People also ask

How much money does cat make?



Is Cat a good investment?



What is cat margin?



Is Cat undervalued?

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Caterpillar Net Income 2010-2022 | CAT - Macrotrends

Caterpillar annual net income for 2019 was \$6.093B, a 0.88% decline from 2018. Compare CAT With Other Stocks ...

Preview for next week: Tolerant retrieval & spelling correction



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<https://insideout.unsw.edu.au>

UNSW - Inside Out

How do your values shape your future? With Inside Out, discover how your values match with different degrees at UNSW and map your future career!



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<https://china.unsw.edu.au> › enginee...

悉尼新南威尔士大学工程学部——全澳第一的 ... - UNSW Sydney

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Contact Us And Meet The Team

Level 2, South Wing, Rupert Myers Building Gate 14, Barker Street UNSW Sydney NSW 2052.

T: +61 2 9385 5555. E: experts@unisearch.com.au. How to get there



chicken celery



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About 73,600,000 results (0.42 seconds)

Recipes :



Stir-fried chicken with celery & shallots

Taste

4.5 ★★★★★ (7)

20 mins

Chicken breast fillets, oyster sauce, soy sauce, white rice,



Spicy Chicken Stir-Fry With Celery and Peanuts

Bon Appetit

4.8 ★★★★★ (253)

Boneless chicken thighs, soy sauce, sambal oelek, dates, rice



Chicken Stir Fry with Celery, Carrot and Cashew

RecipeTin Japan - RecipeTin ...

4.9 ★★★★★ (13)

10 mins

Soy sauce, chicken thigh fillets, sesame oil, cashew nuts, carrot

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People also ask :

How to cook chicken and celery? ▾

Do Chinese use celery in cooking? ▾



chicken OR celery



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Curry

Whole

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Bon appetit

Casserole recip

About 3,340,000,000 results (0.39 seconds)

Recipes :



Chicken recipes

BBC Good Food

No reviews

Chicken wings



Perfect roast chicken

Jamie Oliver

No reviews

1 hr 25 mins

Lemon, carrots, olive oil, garlic,
celery



66 chicken recipes

Olive Magazine

No reviews

Chicken thighs

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People also ask :

How to cook chicken and celery? ▼

Which is the tastiest chicken dish in the world? ▼



chicken AROUND(2) celery



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About 73,600,000 results (0.34 seconds)

Recipes :



Stir-fried chicken with celery & shallots
Taste
4.5 ★★★★★ (7)
20 mins
Chicken breast fillets, oyster sauce, soy sauce, white rice,



Chicken Stir Fry with Celery, Carrot and Cashew
RecipeTin Japan - RecipeTin ...
4.9 ★★★★★ (13)
10 mins
Soy sauce, chicken thigh fillets, sesame oil, cashew nuts, carrot



Chicken Celery Soup
nish kitchen
5.0 ★★★★★ (51)
40 mins
Coconut milk, chicken breast, olive oil, red, chicken stock

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Overview (16) car /s accident

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Cases | View all 10,000

- ☐ **Stanley v. McDonough**
United States Court of Appeals for Veterans Claims. | July 21, 2021 | Not Reported in Vet. App. Rptr. | 2021 WL 3056795

Army veteran Raymond D. Stanley appeals a January 2020 Board decision that denied service connection for a cervical spine (neck) disability and a bilateral knee disability. The issue is whether Mr. Stanley's current cervical spine and bilateral knee disabilities are related to his military service. The Board relied on a 2016 VA exam to conclude...

...at 1084. Third, in 1957, Mr. Stanley was in a **car accident** and incurred "abrasions + contusions" to both knees. R. at 1136...

...that his knee disabilities were caused by an in-service **car accident** and that his neck disability was caused by the snake...
- ☐ **Sadowski v. Tuckpointers Local 52 Health & Welfare Trust**
United States District Court, N.D. Illinois, Eastern Division. | December 20, 2017 | 281 F.Supp.3d 710 | 2017 WL 6549759

LABOR AND EMPLOYMENT - Benefit Plans. ERISA plan administrator's decision to deny medical benefits to participant for removal of her spinal cord stimulator was arbitrary and capricious.

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...implanted two years prior after she was seriously injured in **car accident**. Parties cross-moved for summary judgment. Holding: The District Court...

...paid for spinal cord stimulator to be implanted following participant's **car accident** two years prior and participant could not submit additional bills...

Q&A